Department of Mechanical & Aerospace Engineering
The George Washington University

Guest Talk :

Innovation @ IOT + Human Behaviour + Startup

February 04th, 2026

Anurag Bhatnagar

About the guest :

Shift AltCap
CONNECTING CAPITAL

1998-2017 : As an employee, he launched 5 startups - helped scale 3 of them to unicorns

Since 2017, runs his boutique Investment Bank (*ShiftAltCap*) - work with deep tech / tech enabled startups.

✓ *Mentored, grew their business & raised US $ 40 Mn for 24 startups.*

Also served as a Jury Member for FinTechs and A.I. startups with NASSCOM (*industry association for software companies in India*).

Anurag also runs a portfolio management service offering curated portfolios of securities listed on New York Stock Exchange, where he has delivered to his investor clients an average of ~38% CAGR since 2020.

Practitioner

Flow of the talk :

a) A brief background and context

b) IOT - Key elements, Challenges, Landscape, market and leading players, Key use cases.

c) Human behaviour : vis-a-vis IOTs.

d) What is a problem around you that you can solve using IOT’s ?

e) The confluence : unique opportunities to build valuable businesses.

Practical & Actionable Outcomes

IOT : key elements, Challenges, Landscape, market and leading players, Key use cases.

IOT is the network of interconnected devices embedded with sensors, software, and communication technologies, enabling them to collect, exchange, and act on data over the internet without human intervention.

Elements in the IoT Ecosystem

1. **Devices and Sensors:** Collect data from the environment or users (e.g., smart thermostats, wearables).
2. **Energy & Connectivity:** Networks that power the devices and transmit data (Wi-Fi, Bluetooth, LoRaWAN, 5G).
3. **Computing:**
   a) **Cloud:** Stores and processes vast amounts of data generated by IoT devices.
   b) **Edge:** Process data closer to source, reducing latency.
4. **IoT Platforms:** Provide systems and tools for device management, analytics & integration.
5. **Intelligence**: AI & ML - analyze IoT data for insights, training - and automation to facilitate scalability.
6. **Cybersecurity:** Protects data and devices from unauthorized access.

Key use cases

1. Manufacturing
2. Healthcare
3. Agriculture
4. Smart Homes
5. Retail
6. Logistics
7. Environment Monitoring
8. Energy Management
9. Smart Cities
10. Finance

Leading players

1. Cisco - kinetic platform
2. Bosch - industrial & home
3. IBM - Watson IOT platform
4. Google - NEST, Cloud IoT Core
5. PTC - ThingWorx
6. GE Digital - Predix platform
7. Verizon - 5G and NB-IOT
8. SAP - Business Technology Platform
9. Siemens - Insights Hub, SIMATIC, Mendix
10. Tesla, John Deere - mobility

Challenges

1. Security
2. Interoperability
3. Energy efficiency

Threats

1. Quantum Computing
2. Cybersecurity Risks
3. Regulatory Challenges

Expected : 50 Billion IoT connected devices by 2035 (>*6/human*)

Human behaviour

We think we behave rationally - but Nobel Prize winners Kahneman (2002) and Thaler (2017) demonstrated that humans often make irrational choices. Their theories on behavioural economics have pioneered concepts like ‘nudge’.

‘Nudge’ : alters people's behavior in a predictable way without forbidding any options or significantly changing their economic incentives.

It's about ‘guiding’ (or nudging) choice — not restricting it.

‘Nudge’ in action using IOT

Smart thermostats like **Google Nest** use IoT sensors & learning algorithms to enable homeowners reduce energy usage.
The way they do it is through *nudges*, not mandates.

Default Settings: The thermostat is pre-programmed with energy-efficient temperature settings, but users can change them any time.
⟶ *Nudge: People are more likely to stick with the default.*

Leaf Icon Cue: When users choose energy-efficient temperatures, a green **leaf icon** appears on the screen.
⟶ *Nudge: Provides positive reinforcement, signalling “good behavior” without punishing other choices.*

Monthly Energy Reports: Nest emails users a simple summary comparing their energy usage with neighbours (anonymously), encouraging better habits.
⟶ *Nudge: Social proof makes people want to "do better" than average.*

Behavioural impact:

- Users reduce energy consumption ***without being forced*** to : often saving 10–15% on energy bills.
- The system doesn't penalise inefficient settings but subtly encourages better ones.
- It leverages defaults, gamification (*the leaf*), and social norms to influence choices.

Respects freedom of choice, changes behavior predictably, and is low-cost to implement — all hallmarks of effective nudges.

Startups :

IoT and the inputs it provides in real time are playing an increasingly transformative role in how startups are solving real world problems by enhancing efficiency, improving customer experiences, enabling smarter decision-making and saving lives.

And they are doing it profitably.

Finance	• IOT enabled ATMs - that never run out of cash • Fraud detection - IOT devices plus machine learning identify patterns to detect outliers in real time • Trade Finance - tracking IoT enabled shipments, inventory and production processes in real time • High Frequency Trading (HFT) - that analyse your order and front run it before it gets executed
Mobility	• Tesla has shown a great example of how IOT can save lives • Intelligent traffic lights that change with the density of traffic flow at different hours of the day • Railways use IOT's to signal ahead to unmanned crossings • IOT's are going to be at the core as drone taxi's take off

IOT's are changing the way decisions are made : by humans, or other machines.

The confluence : plus a suggested activity for students

When IOT enabled systems plus principles of ‘nudge’ in human behaviour are combined with a startup mindset, that tri-junction is where the magic of innovation happens.

The Background :

Study the Vision Zero 2026-2030 Integrated Plan committed to by Washington DC - aiming to eliminate traffic fatalities & injuries. One critical hurdle is the lack of data for “near miss” incidents. Currently the District Department of Transportation (DDOT) relies on historical crash data, which has no visibility of these near miss incidents.

Identifying the “near miss” data with the aid of IOT’s can bring visibility to this root cause. And help redesign the infrastructure of the high risk junctures to nudge human behaviour to avoid “near miss” incidents.

Impact : eliminate injuries and make DC safe for pedestrians and drivers alike.

The Project :

GWU students of Department of Mechanical and Aerospace Engineering to design and deploy an **IoT Edge AI system** that transforms un-signalised crosswalks into "intelligent zones" capable of detecting, classifying & reporting near-collisions in real-time without compromising citizen privacy. See separate note co-authored with Professor Bulusu.

The Project can eventually transform into a startup that can be subscribed to by municipal corporations across USA.

The future can be yours to choose : are ***you*** game ?

Thank you for your time !

Anurag Bhatnagar

E : anuragbhatnagar01@gmail.com

M : +91-7600109257